

# The intermediate region as a finite verification: complete analyticity, the transfer matrix, and the two numbers whose meeting closes the proof

The intermediate region has been described in this series as the part of the coupling range where no expansion applies. That is true of expansions, and it leaves out a different class of rigorous tools: the finite-volume criteria for exponential decay of correlations in lattice spin systems, of which the Dobrushin–Shlosman conditions of *complete analyticity* (J. Stat. Phys. 46 (1987) 983) and the strong-mixing conditions of Martinelli and Olivieri (Commun. Math. Phys. 161 (1994) 447 and 487) are the standard forms; all references here are at metadata level. The Wilson measure is a lattice spin system of exactly their type: compact spins  $U_\ell \in SU(3)$ , a product a priori measure (Haar), a translation-invariant finite-range interaction (the plaquette term). Gauge invariance is a symmetry of the interaction and no obstruction, since Elitzur’s theorem (Phys. Rev. D 12 (1975) 3978) only says that gauge-variant local observables have zero expectation. Three consequences, each a chain of known results:

1. If a Dobrushin–Shlosman condition holds for the Wilson interaction at coupling  $\beta_W$ , then truncated correlations of all local observables decay exponentially, uniformly in boundary conditions, and by link reflection positivity the Wilson transfer matrix has a spectral gap: **the theory at  $\beta_W$  is gapped, uniformly in the volume.**
2. The condition is a finite computation on boxes whose side is a few correlation lengths  $\xi/a$ , and the set of interactions satisfying it is open, so an interval of couplings is covered by finitely many verifications, each with an explicit radius.
3. The region where the verification is feasible in principle is the one where  $\xi/a$  is moderate, that is precisely the intermediate region; the weak side, where  $\xi/a = 1/(a\Lambda)$  grows beyond any box, is the domain of the renormalization group, and the target box of the target-box note may be replaced by the verified interval, whose openness radius supplies the tolerance.

So the mass-gap problem for  $SU(3)$  on the lattice reduces to the meeting of two numbers:  $g_{\text{RG}}^2$ , the largest coupling up to which a small-field renormalization argument delivers the effective interaction inside the openness radius of the Wilson interaction; and  $g_{\text{DS}}^2$ , the smallest coupling at which a Dobrushin–Shlosman condition has been verified. The proof closes when  $g_{\text{RG}}^2 \geq g_{\text{DS}}^2$ . Neither number is known today, the verification is a computer-assisted task of a size far beyond present practice, and this repository performs no numerics; what is new here is that the intermediate region is a finite problem with a precise statement rather than a region without tools. Constants where they exist; nothing promoted.

## 1. The Wilson measure as a spin system

Sites: the links of  $\mathbb{Z}^4$  (or of a periodic lattice). Spin space:  $SU(3)$ , compact, with Haar measure  $d\mu$ . Interaction: for each plaquette  $p$ ,

$$\Phi_p(U) = -\frac{\beta_W}{N} \text{Re tr } U_p, \quad \beta_W = \frac{2N}{g^2},$$

a bounded function of the four link variables of  $p$ , translation invariant, of range one. The Gibbs measures in a box  $\Lambda$  with boundary condition  $\bar{U}$  on the links outside  $\Lambda$  are

$$d\mu_{\Lambda}^{\bar{U}}(U) = \frac{1}{Z_{\Lambda}^{\bar{U}}} \exp \left[ - \sum_{p \cap \Lambda \neq \emptyset} \Phi_p(U \vee \bar{U}) \right] \prod_{\ell \in \Lambda} d\mu(U_\ell),$$

well defined for every  $\bar{U}$ , gauge-invariant or not. Gauge transformations at interior sites are a symmetry of each  $\mu_{\Lambda}^{\bar{U}}$  restricted to interior links, so the expectation of a gauge-variant interior

observable vanishes for every  $\bar{U}$  (Elitzur); gauge-invariant observables are the ones with content, and nothing in the Dobrushin–Shlosman framework asks for more than boundedness and finite range of  $\Phi$ .

## 2. The finite-volume criteria and what they imply

Dobrushin and Shlosman give a list of conditions on the finite-volume Gibbs measures, proved equivalent, each of the form: for boxes of a fixed side  $R$  and all pairs of boundary conditions differing at one site, the influence on the interior is small in a specified sense. Any one of them implies: a unique infinite-volume Gibbs state; exponential decay of truncated correlations of all local observables, with a rate  $m > 0$  and uniformly in boundary conditions; analyticity of the free energy and of the expectations of local observables in the interaction. Martinelli and Olivieri prove that a weaker condition, strong mixing on boxes of side  $R$ , suffices for the decay statement in the one-phase region, for general (Part II) as well as attractive (Part I) systems.

**From decay to the gap.** The Wilson action is link-reflection positive, so the transfer matrix  $\mathcal{T}$  in the time direction is self-adjoint and positive,  $H_W = -(\hbar c/a) \log \mathcal{T}$ , and for a local observable  $A$  orthogonal to the vacuum  $\langle A, \mathcal{T}^n A \rangle$  is a truncated Euclidean correlation at time separation  $n$ . Exponential decay at rate  $m$  for all local  $A$  forces the spectral measure of every such  $A$  to vanish above  $e^{-ma}$ , so  $\text{gap}(H_W) \geq \hbar c m$  on the cyclic subspace of local observables, as in the Wilson note §3. Uniformity in the boundary conditions is what makes this hold on every spatial torus.

**Proposition.** If a Dobrushin–Shlosman or Martinelli–Olivieri condition holds for the Wilson interaction at  $\beta_W$ , then the lattice  $SU(3)$  theory at that coupling has a unique vacuum and a spectral gap  $\Delta_W \geq \hbar c m > 0$ , uniformly in the spatial volume.

## 3. Covering an interval, and what “the box” becomes

*Openness.* The set of interactions satisfying a Dobrushin–Shlosman condition on boxes of side  $R$  is open in the norm  $\|\Phi\| = \sup_\ell \sum_{p \ni \ell} \|\Phi_p\|_\infty$ , with a radius that the condition’s own margin fixes. A verification at  $\beta_W^{(i)}$  on side  $R_i$  therefore covers an explicit interval around it, and a finite interval  $[\beta_1, \beta_2]$  is covered by finitely many verifications.

*Feasibility scales with  $\xi/a$ .* The side  $R$  needed is a few correlation lengths in lattice units; at strong coupling  $\xi/a$  is small and the condition is implied by the convergent expansion, while toward weak coupling  $\xi/a = 1/(a\Lambda(g))$  grows like  $e^{1/(2b_0 g^2)}$  and no box suffices. The condition is thus a tool for the intermediate region and only for it.

*The box moves.* Whatever interval  $[\beta_1, \beta_2]$  is verified, the renormalization group of the weak-coupling side no longer has to reach the strong-coupling threshold; it has to deliver an effective interaction within the openness radius of the Wilson interaction at some  $\beta_W \in [\beta_1, \beta_2]$ . The openness radius is the tolerance that the target-box note supplied by hand, and the effective interaction is allowed to contain any finite-range bounded terms, since the criteria are stated for general finite-range interactions.

## 4. The two numbers

Define  $g_{\text{DS}}^2$  as the smallest coupling at which a Dobrushin–Shlosman condition for the Wilson interaction has been verified on some box, and  $g_{\text{RG}}^2$  as the largest coupling up to which a rigorous small-field renormalization argument brings the effective interaction within the openness radius of the Wilson interaction at that coupling. Then:

- for  $g^2 \geq g_{\text{DS}}^2$  the theory is gapped (Section 2);
- for  $g^2 \leq g_{\text{RG}}^2$  the effective theory at some coarser scale lies in the verified region and inherits its gap, provided the renormalization steps are exact low-energy reductions;

- **the mass gap holds along the trajectory if  $g_{\text{RG}}^2 \geq g_{\text{DS}}^2$**  in the refined sense of Section 4b, and T3 then concerns only the continuum limit along the trajectory.

Today  $g_{\text{DS}}^2$  is not below the strong-coupling threshold, no verification having been attempted, and  $g_{\text{RG}}^2$  is not known to exceed zero with an explicit constant. The physical expectation, from the numerical evidence for a smooth crossover with  $\xi/a$  of order 1–10 for  $\beta_W$  between 5 and 6.5, is that a verification at those couplings would need boxes of side 10–30 and that  $g_{\text{DS}}^2$  could in principle be brought to about 1; the verification itself, a rigorous bound on integrals over  $SU(3)^{3R^4}$  with all boundary conditions, is beyond present computational practice by a wide margin, and this repository performs no numerics.

**T3 in this framework.** Suppose the renormalization steps are exact low-energy reductions and the trajectory from bare coupling  $g$  enters the verified interval after  $n(g)$  doublings at an effective interaction  $\Phi_s(g)$ . The gap in lattice units is then  $\delta(g) = 2^{-n(g)} \delta_{\text{eff}}(\Phi_s(g))$ , with  $\delta_{\text{eff}}$  the gap of the verified theory at its own scale, and the ratio  $\delta(g)/(a\Lambda_{\text{lat}}(g))$  of T3 depends on  $g$  only through the point  $\Phi_s(g)$  at which the trajectory enters the interval. Its convergence as  $g \rightarrow 0$  is therefore the statement that the trajectories from different bare couplings converge to one curve as they reach the interval, that is, that the irrelevant directions contract along the flow; the strong-coupling threshold plays no role in it. The clause T3 is in this way absorbed into the same hypothesis on the renormalization map as T2', and the finite verification supplies the gap  $\delta_{\text{eff}}$  with an explicit value at each point of the interval.

#### 4b. Three refinements of the two-number statement

Reviewing the statement “the proof closes when  $g_{\text{RG}}^2 \geq g_{\text{DS}}^2$ ” at full effort gives three corrections, recorded here because each changes what an unblocking would have to look like.

*Sufficient, not necessary.* The meeting of the two reaches is one route, the natural one for any strategy built from expansions at the two ends and a finite check between them. Necessity is not a theorem: a proof of T2' by a different mechanism would bypass the verification altogether. What is true is that every rigorous route known today has this shape, because the confinement scale is a region with no small parameter, and the finite verification is the only general tool that handles such a region.

*The target is a set in interaction space.* The renormalization map does not return the Wilson interaction at a larger coupling; it returns an effective interaction with many terms, exponentially decaying in range. The completely analytical interactions form an open set  $\mathcal{CA}$  in the space of summable interactions, which contains the Wilson interactions at  $g^2 > 444$  by the Dobrushin condition, and the correct statement is: **the trajectory of effective interactions from  $g \rightarrow 0$  must enter  $\mathcal{CA}$ .** The two-number version is the special case in which the trajectory passes close to a Wilson interaction, and “the coupling reaches  $g_{\text{DS}}^2$ ” is neither necessary nor sufficient by itself; the rest of the effective interaction has to lie within the openness radius as well. The entry, if it happens, is at the scale where the effective coupling is of order one, and beyond that scale the block variables decorrelate and the trajectory approaches the trivial fixed point, deep inside  $\mathcal{CA}$ .

*The trajectory statement is weaker than the whole-range statement.* “The lattice theory is gapped for every  $g$ ” is a statement about the Wilson action on its full coupling range, believed true for  $SU(3)$  and false for other actions and groups with bulk transitions. The Jaffe–Witten problem needs only the trajectory from  $g \rightarrow 0$ , that is, the scaling region, and the finite verification is needed only along it. Nothing about couplings the trajectory never visits enters the proof.

*What a hand computation cannot do.* Bounding the block influence  $\rho_V(x, y)$  by the oscillation of the joint log-density, as in Lemma 1 of the Dobrushin note, gives  $e^{4\beta_W} - 1$  for every  $x \in V$  and every boundary link  $y$ , so the block sum is at most  $|V| |\partial V| (e^{4\beta_W} - 1)$  and the block criterion reads  $|\partial V| (e^{4\beta_W} - 1) < 1$ , which is worse than the single-link condition for every  $V$ . Blocks

help only through the decay of  $\rho_V(x, y)$  in the distance from  $x$  to  $y$  inside  $V$ , that is, through screening by the integrated links, and that decay is exactly what the verification computes and what no oscillation bound captures. The single-link threshold  $g^2 > 444$  is therefore the end of what hand bounds of this type give.

## 5. What this changes and what it does not

*Changes.* The intermediate region is a finite verification problem with a precise statement, and the renormalization group’s finish line is an interval with an explicit openness radius rather than the strong-coupling threshold. The three tools of the map, small-field renormalization, finite-volume mixing conditions, and the strong-coupling expansion, cover the coupling range if the first two overlap.

*Does not change.* No step of the proof is completed here. The verification is not performed, the small-field argument is not carried out with constants, and the meeting of  $g_{\text{RG}}^2$  and  $g_{\text{DS}}^2$  is a conjecture supported by numerical evidence outside this repository.

## 6. Consequence for STATE

The intermediate region has a rigorous finite reformulation: verify a Dobrushin–Shlosman condition for the Wilson interaction on finite boxes, transfer the decay to the gap through the transfer matrix, and cover intervals by openness. The proof closes when the renormalization group’s reach  $g_{\text{RG}}^2$  meets the verified reach  $g_{\text{DS}}^2$ . Both are numbers, neither is known, and the second is the one that a computation, outside this repository’s rules, could in principle produce.